Software Project Management (SPM) 🛠️:

- Planning, Resource allocation, Risk management, Scheduling, Quality & Communication
- Ensures on-time, within-budget, quality project delivery

SEI CMM 🏢:

- Maturity model (Levels 1–5: Initial → Optimizing)
- Improves processes, quality, and predictability

ISO-9000 Certification 📜:

- Standards for Quality Management Systems
- Ensures consistent quality, customer satisfaction, process improvement

Software Risk Management ⚠️:

- Identify → Analyze → Prioritize → Mitigate → Monitor risks
- Reduces project failure, cost overruns, improves reliability

CASE Tools 🖥️:

- Upper CASE → Planning & design (Rational Rose, Lucidchart)
- Lower CASE → Coding & testing (Visual Studio, Selenium)
- Integrated CASE → All phases (IBM Rational Suite)
- Benefits: Automation, quality, documentation, productivity

✅ **Tips for Quick Recall:**

- Models → Waterfall 💧, Incremental 🔄, Spiral 🌀
- Quality → CRIMEFUL: Correctness, Reliability, Integrity, Maintainability, Efficiency, Flexibility, Usability, Reusability
- Testing → Top-down 🔝, Bottom-up 🔽
- CASE Tools → Upper, Lower, Integrated

🧠 Software Engineering – 1-Page Memory Map

Unit-1: Basics & SDLC

- **Software** 🖥️ = Programs + Data
- **Software Eng.** 🛠️ = Systematic development & maintenance
- **Characteristics:** CRIMEFUL → Correctness, Reliability, Integrity, Maintainability, Efficiency, Flexibility, Usability, Reusability
- **Applications:** System, App, Web, Mobile, Embedded, Scientific, Business
- **SDLC Phases:** Requirement → Design → Coding → Testing → Deployment → Maintenance
- **Models:**
  - Waterfall 💧 → Linear, sequential
  - Incremental 🔄 → Build in parts
  - Spiral 🌀 → Iterative, risk-driven
- **Quality Factors:** Correctness, Reliability, Efficiency, Integrity, Usability, Maintainability, Portability, Reusability, Flexibility

Unit-2: Requirements & Patterns

- **SRS** 📄 → Correct, Complete, Clear, Consistent, Verifiable, Modifiable, Traceable
- **Project Management** 🛠️ → Planning, Scheduling, Resources, Risk, Monitoring, Quality, Communication
- **FO vs OO:**
  - FO → Functions, separate data, procedural (C)
  - OO → Objects (data+methods), reusable, real-world (Java)
- **Design Patterns** 🧩:
  - Creational → Singleton, Factory
  - Structural → Adapter, Decorator
  - Behavioral → Observer, Strategy

Unit-3: Programming & Testing

- **Structured Programming** 🏗️ → Sequence, Selection, Iteration, Modular, Top-down
- **Information Hiding** 🛡️ → Encapsulation + Abstraction + Access Modifiers
- **Coding Styles** 💻 → Modular, Structured, OOP, Declarative, Scripting
- **Testing:**
  - Top-down 🔝 → test high-level first, stubs
  - Bottom-up 🔽 → test low-level first, drivers
- **Test Cases & Criteria** 🧪 → Steps/Input/Expected/Actual + Standards (Functional, Performance, Reliability, Security)

Unit-4: Management & Tools

- **COCOMO** 🏗️ → Effort estimation from LOC (Basic, Intermediate, Detailed)
- **SPM** 🛠️ → Plan, Schedule, Resources, Risk, Quality, Communication
- **SEI CMM** 🏢 → Levels 1–5: Initial → Optimizing → Process maturity
- **ISO-9000** 📜 → Quality standards, customer focus, continuous improvement
- **Software Risk Mgmt** ⚠️ → Identify → Analyze → Prioritize → Mitigate → Monitor
- **CASE Tools** 🖥️ → Upper (Design), Lower (Coding/Testing), Integrated (All phases)

🔹 Quick Recall Tips:

- Models → 💧 🔄 🌀
- Quality → CRIMEFUL
- Testing → 🔝 Top-down, 🔽 Bottom-up
- Design Patterns → Creational, Structural, Behavioral 🧩

🧩 Software Engineering – Emoji Mind Map

Unit-1: Basics & SDLC

scss

```
Software 🖥️
│
├─ Software Eng. 🛠️
│    └─ Systematic Dev + Maintenance
│
├─ Characteristics CRIMEFUL ✨
│    └─ Correctness ✔ Reliability ✔ Integrity ✔ Maintainability ✔ Efficiency
✔ Flexibility ✔ Usability ✔ Reusability ✔
│
├─ Applications 📂
│    ├─ System 💻
│    ├─ App 📱
│    ├─ Web 🌐
│    ├─ Mobile 📱
│    ├─ Embedded 🔌
│    ├─ Scientific 🧬
│    └─ Business 💼
│
├─ SDLC 🌀
│    └─ Requirement → Design → Coding → Testing → Deployment → Maintenance
│
└─ Models
     ├─ Waterfall 💧 (Linear)
     ├─ Incremental 🔄 (Parts)
     └─ Spiral 🌀 (Iterative, Risk)
```

Unit-2: Requirements & Patterns

```scss
SRS 📄
│
├─ Good SRS ✔ Correct, Complete, Clear, Consistent, Verifiable, Modifiable,
Traceable
│
Project Mgmt 🛠️
│
├─ Planning 📋
├─ Scheduling ⏰
├─ Resources 👥💻
├─ Risk ⚠️
├─ Monitoring 📊
├─ Quality ✅
└─ Communication 💬
│
Approaches
├─ FO → Functions 🔹 Data separate
├─ OO → Objects 🟢 Data+Methods
│
Design Patterns 🧩
├─ Creational 🔨 Singleton, Factory
├─ Structural 🏗️ Adapter, Decorator
└─ Behavioral 🔄 Observer, Strategy
```

Unit-3: Programming & Testing

```mathematica
Structured Programming 🏗️
│
├─ Sequence → Selection → Iteration
├─ Modular ✅ Top-down
│
Information Hiding 🛡️
├─ Encapsulation + Abstraction + Access Modifiers
│
Coding Styles 💻
├─ Modular
├─ Structured
├─ OOP
├─ Declarative 📝
└─ Scripting 🐍
│
Testing 🔬
├─ Top-down 🔝 (High → Low, Stubs)
├─ Bottom-up 🔽 (Low → High, Drivers)
└─ Test Cases 🧪 Steps/Input/Expected/Actual + Criteria (Functional,
Performance, Reliability, Security)
```

Unit-4: Management & Tools

```sql
COCOMO 🏗️ → Effort Estimation (LOC)
SPM 🛠️ → Plan, Schedule, Resources, Risk, Quality, Communication
SEI CMM 🏢 → Levels 1–5 Initial → Optimizing
ISO-9000 📜 → Quality Standard, Customer Focus, Continuous Improvement
Risk Management ⚠️ → Identify → Analyze → Prioritize → Mitigate → Monitor
CASE Tools 🖥️
├─ Upper (Design) 🖊️
├─ Lower (Coding/Testing) 💻
└─ Integrated (All Phases) 🌐
```

🔹 Quick Recall

- Models → 💧 🔄 🌀
- Quality → CRIMEFUL ✨
- Testing → 🔝 / 🔽
- Design Patterns → Creational / Structural / Behavioral 🧩